

PROVAKAR MONDAL

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EDUCATION

Virginia Tech

PhD in Computer Science. Advisor: [Dr. Eli Tilevich](#)

Blacksburg, VA, USA

August 2021 – December 2026

Dissertation: Middleware for Robust Integration, Testing, and Error Recovery of Replicated Data Systems

Bangladesh University of Engineering and Technology

BS in Computer Science

Dhaka, Bangladesh

February 2015 – April 2019

WORK EXPERIENCE

Virginia Tech, Software Innovations Lab

Graduate Research Assistant

Blacksburg, VA, USA

August 2021 – Present

- Creating a systematic integration testing framework for agentic AI workflow orchestration, leveraging domain-aware constraints to make exhaustive testing feasible (*ongoing*).
- Provided comprehensive integration testing support for replicated data systems by enabling 32% more test executions within the same time budget via four novel search-space pruning algorithms. [🔗 Code](#)
- Automated architectural refactoring for Java Spring Boot microservices by consolidating distributed components into modular monoliths, reducing invocation latency (65%) and memory consumption (72%). [🔗 Code](#)
- Increased reliability of replicated data systems by 25% under erroneous updates by creating middleware that traces causal, distributed effects of updates and performs recovery operations non-intrusively. [🔗 Code](#)
- Improved cross-language interoperability in replicated data systems via a language-agnostic interface for polyglot replica coordination, achieving 54% performance gain and 38% better software quality. [🔗 Code](#)

Uber, Airports and Travel Team

PhD Software Engineer Intern

Seattle, WA, USA

Summer 2026

- Enhanced a single-passenger, real-time trip-dispatch to support multi-rider trips at airports, enabling drivers finishing a trip to be proactively offered a follow-up trip instead of returning to a holding queue.
- Proposed the architectural changes to support the new dispatch, implementing it via 8 production changes spanning 2 backend services and 4 distinct pipeline stages.
- Devised 48 unit test scenarios for every combination of the feature's conditions, achieving 98.7% test coverage; carried out an end-to-end test plan across driver and rider clients to validate the full pipeline pre-launch.
- Rolled the feature out as a canary release to a target market, using traffic shadowing at each pipeline stage to confirm zero regression to existing trips before enabling.

Samsung Research, Mobile Application Group

Software Engineer

Dhaka, Bangladesh

May 2019 – July 2021

- Contributed to the successful release of *Samsung Notes* v2→v3 by implementing six new features and fixing 100+ bugs, improving app stability and user experience.
- Engineered a delta-data transferable middleware for *Samsung Notes*, reducing synchronization latency by 27% between *Notes* and *Samsung Cloud*, enhancing data consistency and accessibility across devices.

TECHNICAL SKILLS

- **Languages**: Python, Go, JavaScript, C++, Java, C#, C, XAML, HTML, CSS, Protobuf, Prolog
- **DBMS**: Oracle, MySQL, NoSQL, MongoDB, SQLite
- **Frameworks and Libraries**: Node.js, React, Django, UWP, .NET, NumPy, Matplotlib, TensorFlow, PyTorch
- **DevOps, Deployment, and AI Workflow Platforms**: Git, Perforce, Docker, Vercel, Langflow

PUBLICATIONS

- Provakar Mondal and Eli Tilevich. “Automated and Configurable Undo for CRDT-Based Distributed Systems via Metaprogramming” in *Journal of Systems and Software (JSS)* 2026. [📄 Paper](#) [🔗 Code](#)

- **Provakar Mondal** and Eli Tilevich. “Cross-Platform Integration and Extension of Replicated Data Libraries in Distributed Systems” in **Journal of Parallel and Distributed Computing (JPDC)** 2026. [📄 Paper](#) [🔗 Code](#)
- **Provakar Mondal** and Eli Tilevich. “ER- π : Exhaustive Interleaving Replay for Testing Replicated Data Library Integration” in **Middleware 2025**. [📄 Paper](#) [🔗 Code](#)
- **Provakar Mondal**. “Toward Thorough and Practical Integration Testing of Replicated Data Systems” in **Middleware 2025: Doctoral Symposium**. [📄 Paper](#) [🔗 Code](#)
- **Provakar Mondal** and Eli Tilevich. “Understanding Tradeoffs of Replicated Data Library Integration Strategies in Multilingual Environments” in **Middleware 2025: Demos & Posters**. [📄 Paper](#) [🔗 Code](#)
- **Provakar Mondal** and Eli Tilevich. “Undoing CRDT Operations Automatically” in **International Conference on Cloud Computing Technology and Science (CloudCom)** 2023. [📄 Paper](#) [🔗 Code](#)

SELECTED ENGINEERING PROJECTS ---

- **Reliable Replicated Data Systems via Trust Management**: Enhanced fault tolerance by 21% under malicious updates by integrating dynamic trust management and service-level agreement. [🔗 Code](#)
- **AUTOPROB4JAVA**: Integrated probabilistic reasoning into Java applications, reducing coding effort (22%) and boosting query efficiency (43%) via auto-generated logic programs from Java annotations. [🔗 Code](#)
- **COVID-19 Twitter Misinformation Lookup**: Identified COVID-19-related misinformation tweets with 71.56% precision by implementing an Information Retrieval (IR) model using RankNet. [🔗 Code](#)
- **Online Flea Market**: Developed an e-commerce web application for trading pre-owned goods, validated by an in-class user study with 50+ participants, emphasizing interface usability and design quality. [🔗 Code](#)

ACADEMIC SERVICE ---

- **Graduate Application Developer, TLOS, Virginia Tech**: Architected and deployed agentic workflows and RAG pipelines to enable an AI-powered Canvas course enhancement system (August 2025 – May 2026).
- **Adjunct Instructor, Virginia Tech**: Taught Intermediate Programming in Python (Summer 2025).
Topics Covered: Data Science, TDD, OOP, Web Data, Web Scraping, Plotting, ML.
- **Graduate Teaching Assistant, Virginia Tech**: Data Structures & Algorithms, Programming Language Theory & Practice, Comparative Languages, Computer Organization (Fall’21, 22, 24 & Spring’22, 25).
- **External Program Committee**: ICWS 2025, ICCCN 2025, and SPLASH-E 2023.
- **Judge**: Capstone/Industry Category, VTURCS 2026.

AWARDS ---

- **NSF Travel Grant**: Middleware 2025 Conference
- **BitShares Fellowship**: Department of Computer Science, Virginia Tech (2024 – 2025)